

5 minutes quiz

1. Find the following amounts:

a) 10% of \$350 b) 2% of \$600 c) 6% of \$300

$$a) 10\% \text{ of } \$350 = \$35$$

$$b) 1\% \text{ of } \$600 = \$6$$

$$2\% \text{ of } \$600 = \$12$$

$$c) 6\% \text{ of } \$300$$

$$1\% \text{ of } \$300 = \$3$$

$$6\% \text{ of } \$300 = \$3 \times 6 \\ = \$18$$

2. The following students gained these percentages on a Year 11 test. If the test was out of 60 marks, calculate each student's score to the nearest whole number:

a) Cameron 55% b) Janelle 97%

$$\begin{aligned} \text{a) } 50\% \text{ of } 60 &= 30 \\ 5\% \text{ of } 60 &= 3 \end{aligned}$$

Cameron's score = 33 marks

$$\begin{aligned} \text{b) } 1\% \text{ of } 60 &= 0.6 \\ 3\% \text{ of } 60 &= 3 \times 0.6 = 1.8 \\ \text{Therefore } 97\% \text{ of } 60 &= 60 - 1.8 \\ &= 58.2 \end{aligned}$$

Janelle's mark is 58 mark

$$\begin{array}{r} \text{or} \\ 90\% \text{ of } 60 = 54 \\ 7\% \text{ of } 60 = 4.2 \\ \hline 58.2 \end{array}$$

Janelle's mark is 58 mark

3. The price of a commodity increases from \$1 640 per tonne to \$1 846 per tonne. Express the increase as a percentage increase.

$$\% \text{ increase} = \frac{\text{change or difference}}{\text{original}} \times 100\%$$

$$\begin{aligned} \text{change or difference} &= 1846 - 1640 \\ &= \$206 \end{aligned}$$

$$\begin{aligned} \% \text{ increase} &= \frac{206}{1640} \times 100\% \\ &= 12.5609\% \\ &\approx 12.56\% \end{aligned}$$

Consumer Arithmetic - Percentages

Application of percentages

- **Discount**
- **Inflation**
- **Profit and loss**
- **GST**
- Commission
- Taxes
- **Interest – simple and compound**
- **Others**

Discount

- Reduction of price
- Reasons:
 - Sale
 - Bulk buy
 - Damaged goods
 - Cash purchases
 - Customer loyalty
- Usually express in % of the original price



$$\text{Discount} = \text{original price} * \% \text{ discount}$$

$$\text{Discounted price} = \text{original price} - \text{discount}$$

- Company X is giving a discount of 15% for all their computers. What is the discounted price for a computer with a price tag of \$2 100.

Method 1

Calculate discount

$$\begin{array}{r} \text{Discount: } 10\% \text{ of } \$2100 = \$210 \\ 5\% \text{ of } \$2100 = \$105 \\ \hline 15\% \text{ of } \$2100 = \$315 \end{array}$$

$$\$2100 - \$315 = \$1785$$

The discounted price is \$1785

Method 2

$$\text{Discount} = 15\%$$

The new price will be $1 - 15\%$ of the original price, i.e. 85%

$$10\% \text{ of } \$2100 = \$210$$

$$80\% \text{ of } \$2100 = \$2100 \times 8 = \$1680$$

$$5\% \text{ of } \$2100 = \$105$$

The discounted price is
\$1 785

- Mr Smith increases the price of the furniture he sells by 5%. If the original price of a cupboard is \$1 200, what is the new price?

Method 1

$$10\% \text{ of } \$1200 = \$120$$

$$5\% \text{ of } \$1200 = \$60$$

$$\begin{aligned}\text{The new price} &= \$1200 + \$60 \\ &= \$1260\end{aligned}$$

The new price of a cupboard is \$1260

Method 2

$$\begin{aligned}\text{The increase will be } &1 + 0.05 \\ &= 1.05\end{aligned}$$

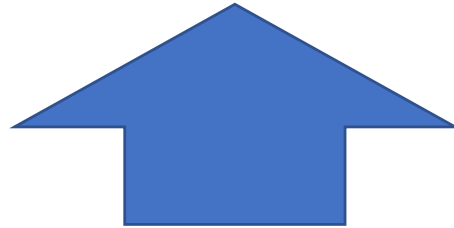
5%
↓

The new price is

$$\$1200 \times 1.05 = \$1260$$

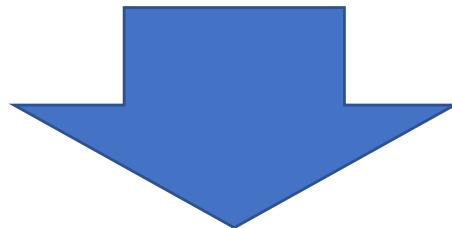
Final Amount

Multiply by $1 + \frac{a}{100}$



Increase by a%

Original Amount,
X



Decrease by a%

Final Amount

Multiply by $1 - \frac{a}{100}$

- After discount of 20%, the price of a phone is \$400.

What is the original price?

① Let x = original price

$$80\% x = 400$$

$$x = \frac{400}{80\%}$$

$$= \frac{400}{80} \times 100$$

$$= \$500$$

The original price is \$500

②

$$80\% \text{ — } \$400$$

$$1\% \text{ — } \frac{400}{80} = 5$$

$$100\% = 5 \times 100 = \$500$$

The original price is \$500

- After inflation (3% p.a), the price of a car is \$21 000.

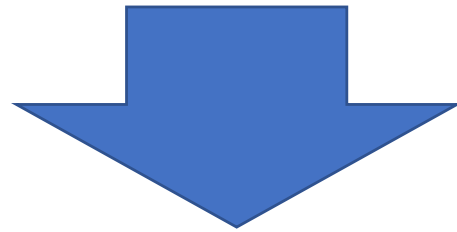
What was the price last year?

$$\$21\,000 \rightarrow 1.03 \text{ of the price last year}$$

$$\text{The price last year} = \frac{21000}{1.03} = 20388.34951$$

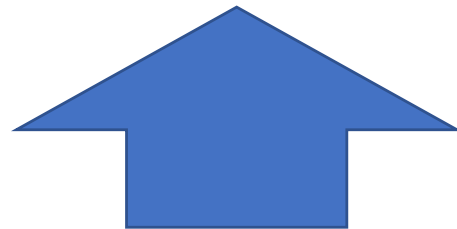
$$\text{The price last year} = \$20388.35$$

Amount increased by a%



Divide by $1 + \frac{a}{100}$

Original Amount,
X

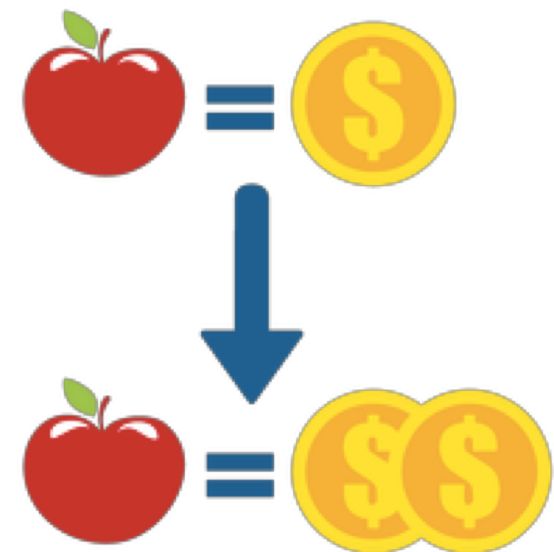


Divide by $1 - \frac{a}{100}$

Amount decreased by a%

Inflation

- Rise in the price of goods over a period of time
- Usually in annual percentage rate
- Different countries have different inflation rates
- Link to purchasing power





1994



2004



2014

- Assume that the annual rate of inflation is 3%, and the current price of a particular item is \$54. What is the price of that item in a) one year; b) 3 years?

$$\text{Inflation} = \text{original price} \times (1 + a\%)^n$$

a% - inflation n - no. of year

$$\text{Inflation} = 3\%$$

$$\text{a) } \$54 \times 1.03$$

$$= \$55.62$$

$$\text{b) } \$54 \times (1.03)^3$$

$$= \$59.01$$

Goods and Services Tax (GST)

- Goods and services tax (GST) is a broad-based tax of 10% on most goods, services and other items sold or consumed in Australia. (Australian Tax Office)
- Rate – 10% in Australia
- Different in other countries
- Called Value Added Tax (VAT) in some other countries



Calculating Price including GST

$$\begin{aligned}\text{Price including GST} &= \text{original price} + \text{GST} \\ &= \text{original price} + (\text{original price} * 10\%) \\ &= \text{original price} * (1 + 10\%) \\ &= 1.1 * \text{original price}\end{aligned}$$

Calculating GST

$$\text{GST} = \frac{\text{Price including GST}}{1.1}$$

Profit and loss

- Profit – Selling something above the price we paid for
- Loss – Selling something below the price we paid for



Price we paid for the item = $\$x$

Price we sold of the item = $\$y$

sold - paid

Difference in price = $\$y - \x

If $\$y - \$x > 0$, we made a profit

If $\$y - \$x < 0$, we made a loss

If $\$y - \$x = 0$, we broke even

Percentage profit/loss

$$\text{Percentage profit} = \frac{\text{profit made when you sell the item}}{\text{amount you paid for the item}} * 100\%$$

$$\text{Percentage loss} = \frac{\text{loss made when you sell the item}}{\text{amount you paid for the item}} * 100\%$$